

QA Process

APPN Aerial SOP — Locked revision 1.0

APPN – Aerial Data QC Protocols

Note

This document describes procedures for measuring the uncertainty of drone flights. Initially this document focuses on the hyperspectral drones, but the procedure will expand in the future. It will also form the basis of a standard QC procedure for future flights.

Note

The QC procedures in this document are tightly coupled to the APPN flight designs — the panels, ground control points, and reflectance targets that the QC steps rely on are deployed *during* the flight itself. Before performing any QC, make sure you understand which targets were placed in the field by reading the relevant flight protocol:

- Standard Flight — routine APPN data-collection flight; defines the baseline panel and GCP layout used for ELM and positional QC.
- Validation Flight — extended flight that adds additional validation panels, GCPs, and (in future) LiDAR calibration targets used by the spectral, positional, and LiDAR QC sections below.

Document Structure

The conventions in General QC Conventions apply to **all** QC processes below. Read that section first, then jump to the specific QC type you need:

- General QC Conventions
 - File Format — GeoJSON vs Shapefile
 - Naming Conventions
 - File Storage
 - Positional QC
 - Spectral QC
 - Creating Geospatial Vector Data — GOBI & CALViS (QGIS)
 - Extracting Pixels into a Table
 - LiDAR QC
-

General QC Conventions

These file-format, naming, and storage rules apply to every QC process (positional, spectral, LiDAR) described later in this document.

File Format — GeoJSON vs Shapefile

Important

GeoJSON (.geojson) is the preferred format for all QC vector data, though shapefiles (.shp) are also supported by the QA code at https://github.com/ArdenB/APPN_GenricFileStorage.

GeoJSON is preferred because:

- It is a **single self-contained file**, rather than the 4-6 sidecar files (.shp, .shx, .dbf, .prj, .cpg, ...) required by the shapefile format. This makes it easier to copy, share, and store without losing pieces.
- It is **plain-text JSON**, so it is human-readable, diff-friendly, and works cleanly with Git version control.
- It is an **open, web-native standard** (RFC 7946) supported by virtually all modern GIS tools, web mappers, and Python/R libraries.
- CRS handling is explicit: the default GeoJSON specification (RFC 7946) only supports WGS84 (EPSG:4326), but all major GIS software (QGIS, ArcGIS, GDAL, geopandas, etc.) allow you to specify and read a different CRS when writing/reading the file. For APPN QC work we **keep the file in the same projected CRS as the source orthomosaic** (e.g. GDA2020 / MGA zone XX) rather than reprojecting to WGS84.
- It has **no field-name length limit** (shapefile DBF caps at 10 characters) and no 2 GB file-size cap.

If you already have shapefiles, you do **not** need to re-create them — the QA pipeline will read either format. New files should be saved as **.geojson**.

Naming Conventions

Name	Overview
QC_ELM_Panels.geojson	Polygons of the reflectance panels used in the ELM during GRYFN Processing.
QC_VAL_Grfyn_Panels.geojson	When a second set of GRYFN panels is placed in the field. Replace <i>ELM</i> with <i>VAL</i> for validation.
QC_VAL_{PanelName}_Panels.geojson	Any future validation panels or tests of other panels. Replace {PanelName} with the name or unique identifier.
QC_GCP_points.geojson	If alternative GCP points (e.g. Aeropoints) are placed in the field. Points-only file; points should match panel centre
QC_LIDAR_{TargetName}_Surface.geojson	Name for any future LiDAR calibration surfaces.

Tip

Any additional information (e.g. date) can be added to the end of the file name with an underscore — e.g. **QC_ELM_Panels_20260302.geojson**. This table will be updated if we source panels other than GRYFN. Shapefile equivalents (e.g. **QC_ELM_Panels.shp**) are also accepted by the QA code.

File Storage

Any files created in the quality control steps are to be stored in a newly created **QC_data** folder inside **T1_proc**, following the APPN folder structure.

Formal path:

```
./{Node}/  
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/  
{YYYYSiteName[_F|C]}/  
{SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/QC_data/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/CALVIS/20250825/run_00/T1_proc/QC_data/
```

These vector files (GeoJSON preferred, shapefile accepted) are meant to be created after the GPRO has been completed. The APPN storage repo has been updated to include this folder: https://github.com/ArdenB/APPN_GenricFileStorage

Positional QC

Warning

This section is still a work in progress.

Important

Positional QC **must be performed immediately after processing is complete** (e.g. straight after GPRO creation). Positional errors are often the first indication of a larger problem with the flight or the processing pipeline, and almost always require the data to be reprocessed before they can be resolved. Catching them early avoids wasted downstream work.

APPN Positional QC is performed in three stages:

1. **Field Data Collection**
2. **Observed point capture**
3. **Accuracy reporting**

The descriptions below show the procedure for the CALViS using GCPs. The process for the GOBI is the same, but there is only the VNIR orthomosaic. The same workflow can be used to produce shapefiles if preferred — just choose *ESRI Shapefile* in place of *GeoJSON* in the format dropdown.

For the CALViS, in the products folder of the completed GPRO you will find the `.tif` files:

- `X_VNIR_Orthomosaic.rgb`
- `X_SWIR_Orthomosaic.rgb`

These are the RGB bands from the VNIR and SWIR files respectively. They can be easier to use than the full `.bin` files, though the procedure is the same.

Field Data Collection

Important

TODO: Name and details pending.

Independent GCPs are placed in the field during data capture (see Standard Flight and Validation Flight). These GCPs are independent of any points used during processing so they provide an unbiased check.

The raw data from the GCP points should be saved in the `T0_raw/Vault` folder (location may change).

Formal path:

```
./{Node}/  
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/  
{YYYYSiteName[_F|C]}/  
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/Vault/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson/CALVIS/20250825/run_00/T0_raw/Vault/
```

This data should then be converted into the APPN standard format (geojson with ID, X, Y, Z, CRS). The exact process for doing this will depend on the exact GCP used (Aeropoint, Trimble, etc).

Important

TODO: Document the conversion process for each supported GCP type (Aeropoint, Trimble, ...).

There are some common issues:

- mismatched CRS between GCP collection and data processing.

- incorrect height data format e.g. GDA2020 ellipsoid and the Australian Height Datum (AHD) — see Geoscience Australia's AUSGeoid2020 conversion tool for converting between ellipsoidal and AHD heights.
- inconsistent ID names or duplicate points. Fix and remove.

Once all the issues are fixed. The document should be saved as __TODO:

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/QC_data/QC_TODO:.geojson
```

Example:

```
./USYD_Narrabri/2025_SIFCa1/2025IAWatson/CALVIS/20250825/run_00/T1_proc/QC_data/QC_TODO:.geojson
```

Observed point capture

Manually digitise a matched set of points from the drone orthomosaic in QGIS and save them as QC_GCP_points.geojson (see Naming Conventions). The GCP_name column must match the names used in the field data.

1. Load the Data

1. Load the SWIR and VNIR files into QGIS and run the following checks:

- Do the panels in the SWIR and VNIR overlap?
- Make note of the CRS.

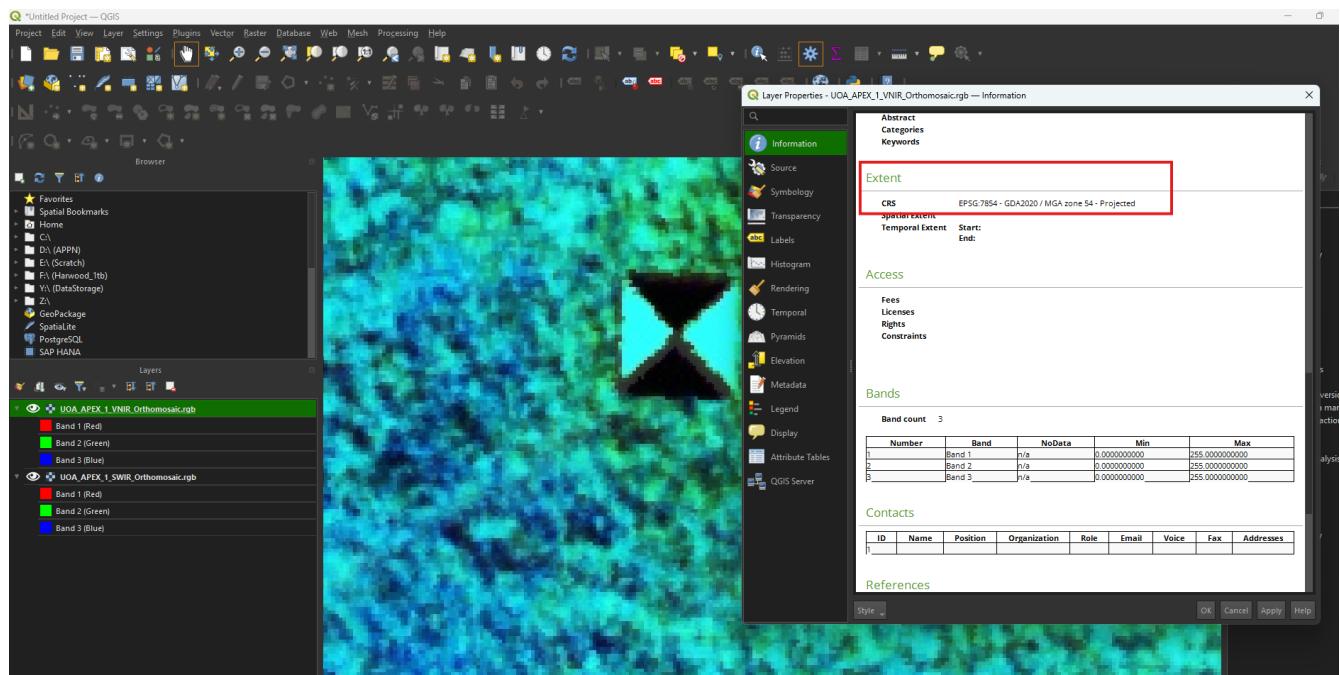
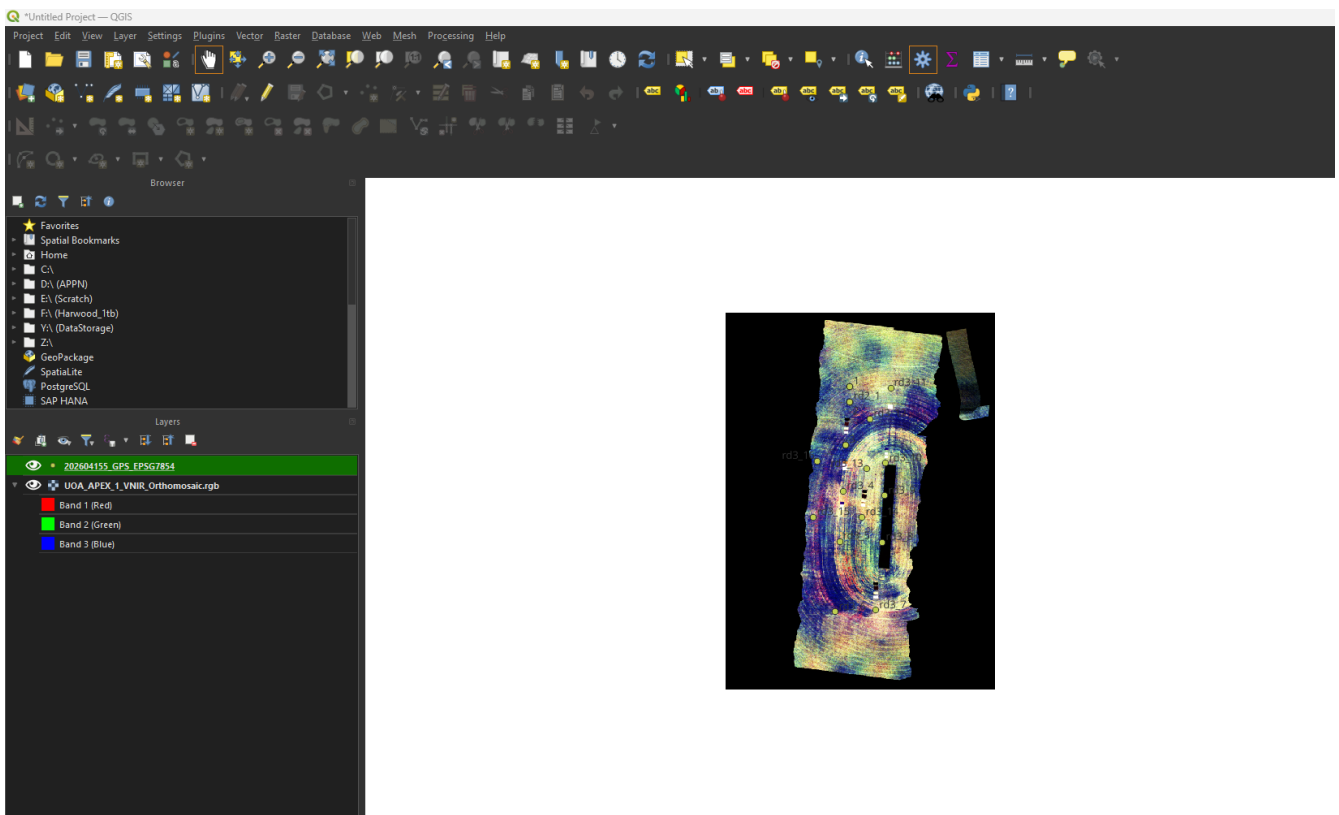


Figure: The VNIR is set to 50% opacity to confirm the panels overlap with the SWIR. The information panel is open on the SWIR to confirm the CRS (EPSG:7854 – GDA2020 / MGA zone 54). This CRS is for Roseworthy SA where this CALViS flight was collected.

Instead of transparency you can also turn one image on and off to confirm overlap is acceptable.

If overlap is fine:

2. Load the ground truth GPS data for your GCPs. Note that the ground truth data is loaded to making naming the annotations on the GRYFN data easier.



Here, is an example of the ground truth data points over the GCP from the CALVIS



2. Create the Shapefile

1. Navigate to **Layer** → **Create Layer** → **New Shapefile Layer**.

The file name will be QC_GCP_points, Geometry type is "points" and for field we delete the existing "ID" and replace it with GCP_name (Note: use Type "Text(string)")

New Shapefile Layer

File name: QC_GCP_points.shp

File encoding: UTF-8

Geometry type: Point

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

Project CRS: EPSG:7854 - GDA2020 / MGA zone 54

New Field

Name:

Type: Text (string)

Length: 80 Precision:

Add to Fields List

Fields List

Name	Type	Length	Precision
GCP_name	String	80	

Remove Field

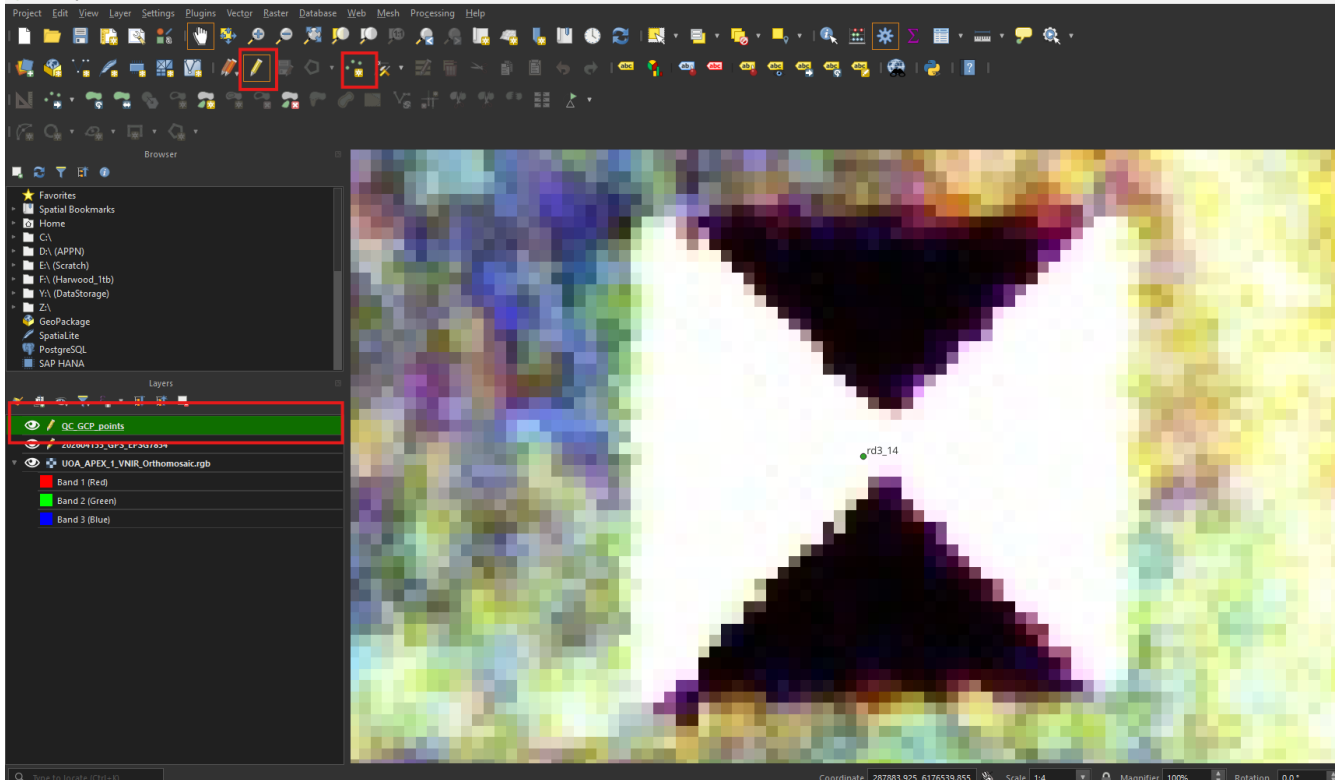
OK Cancel Help

- Set the **File Name** to QC_GCP_points.shp
- Set the **Geometry type** to *points*.
- Set the **CRS** to match your dataset.
- Add a single field — GCP_name:

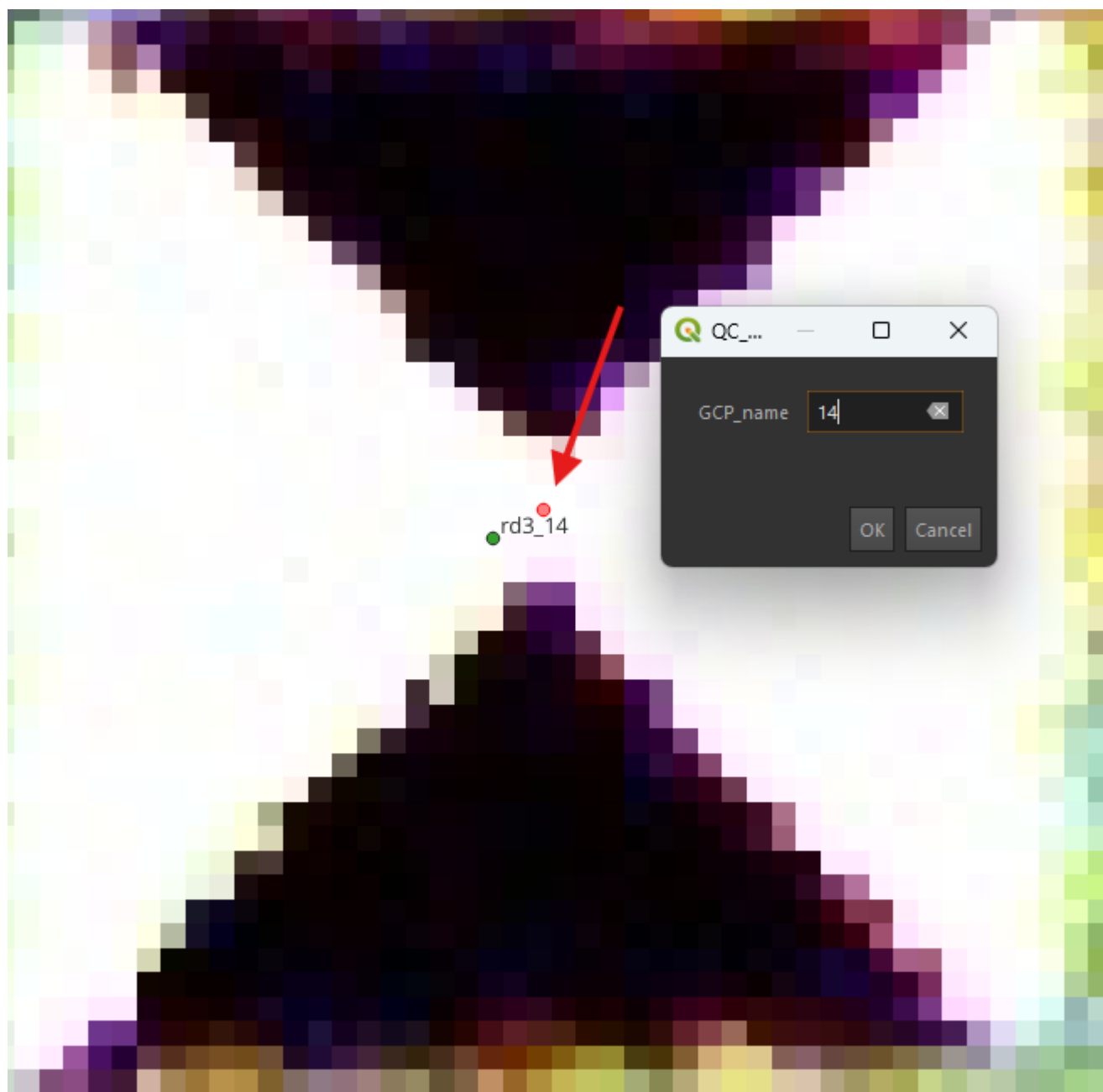
- Select the pre-filled `id` field in the Fields list and click **Remove Field** (bottom right) to remove it.
- Set `GCP_name` as an **Text(string)**

3. Annotating the GCPs To start annotating the GRYFN data:

1. Select `QC_GCP_points` in the **Layers** menu.
2. Click the pencil icon (**Toggle Editing**).
3. Click the points (**Add point feature**)



4. Navigate to the centre of the GCP, left click and name the GCP appropriately (the name needs to match your ground truth data). In the figure below the ground truth data is in Green and the Red dot is GRYFNS accuracy.



Do this for all relevant GCPs.

5. Save the data as a GeoJSON

1. Right-click **QC_GCP_points** in the **Layers** menu → **Export** → **Save Features As**.
2. The presets are fine — just make sure the **File Name** is correct and in the right folder (click the three dots to navigate). Make sure **Format** is GeoJSON and double-check the **CRS**.

Save Vector Layer as...

Format: GeoJSON

File name: D:\GPT_LOCAL\20260415\run_00\T1_proc\QC_data\QC_GCP_points.geojson

Layer name:

CRS: EPSG:7854 - GDA2020 / MGA zone 54

Encoding: UTF-8

☐ Save only selected features

▶ Select fields to export and their export options

☒ Persist layer metadata

▼ Geometry

Geometry type: Automatic

☐ Include z-dimension

☒ Force multi-type

▶ ☐ Extent (current: none)

▼ Layer Options

COORDINATE_PRECISION: 15

RFC7946: NO

WRITE_BBOX: NO

▶ Custom Options

☒ Add saved file to map

OK Cancel Help

Accuracy reporting

Run the QA code at https://github.com/ArdenB/APPN_GenricFileStorage (Code/DS02_DatasetQA/) to generate a spatial accuracy report comparing the observed (drone) and reference (surveyed) GCP locations. The report produces per-point residuals and overall RMSE in X, Y, and Z.

Important

TODO: Write this section and include lots of info.

Spectral QC

Spectral QC is done by drawing polygons in GIS software over surfaces with known spectral values, then extracting and comparing them. This is done after all processing in software like GPT is complete.

CALViS and GOBI flights conducted prior to the *Operational Excellence in APPN Hyperspectral Imaging* SIF likely consist of one GRYFN reflectance panel (used to generate the ELM in the GRYFN Processing Tool) along with additional panels for validation.

File naming, format (GeoJSON preferred), and storage location for the polygons created below all follow the General QC Conventions above.

Creating Geospatial Vector Data — GOBI & CALViS (QGIS)

The description below shows the procedure for creating GeoJSON files for the CALViS using GRYFN reflectance panels for ELM. The process for the GOBI is the same, but there is only the VNIR orthomosaic. The same workflow can be used to produce shapefiles if preferred — just choose *ESRI Shapefile* in place of *GeoJSON* in the format dropdown.

For the CALViS, in the products folder of the completed GPRO you will find the .tif files:

- X_VNIR_Orthomosaic.rgb
- X_calvis_sif_cal_20250925_SWIR_Orthomosaic.rgb

These are the RGB bands from the VNIR and SWIR files respectively. They can be easier to use than the full .bin files, though the procedure is the same.

1. Load the Data

1. Load both files into QGIS and run the following checks:

- Do the panels in the SWIR and VNIR overlap?
- Make note of the CRS.

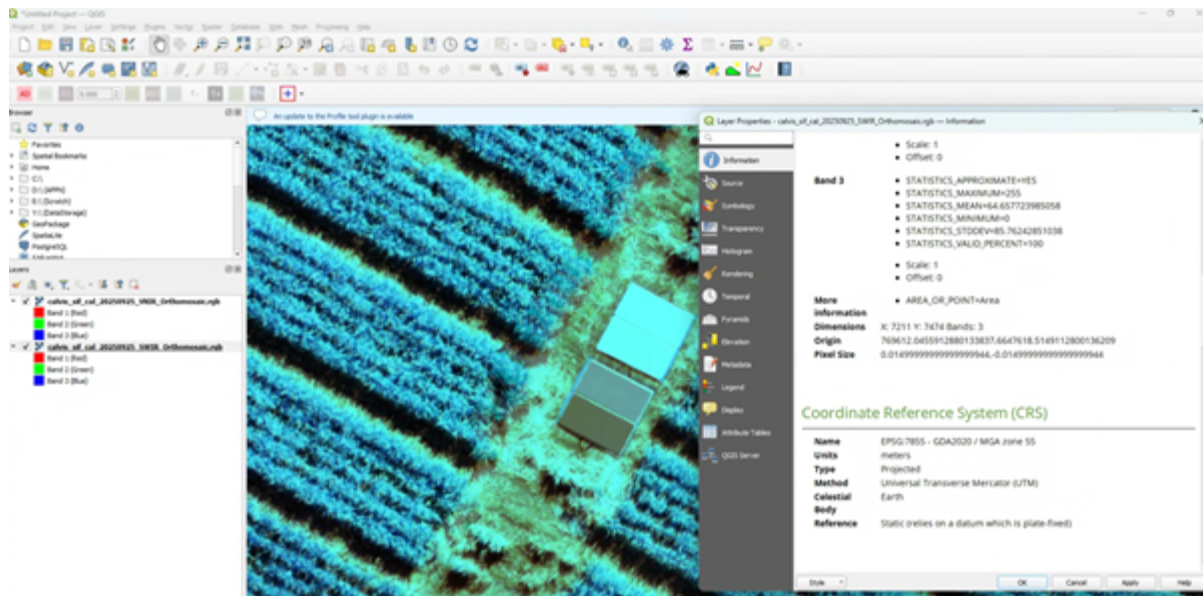
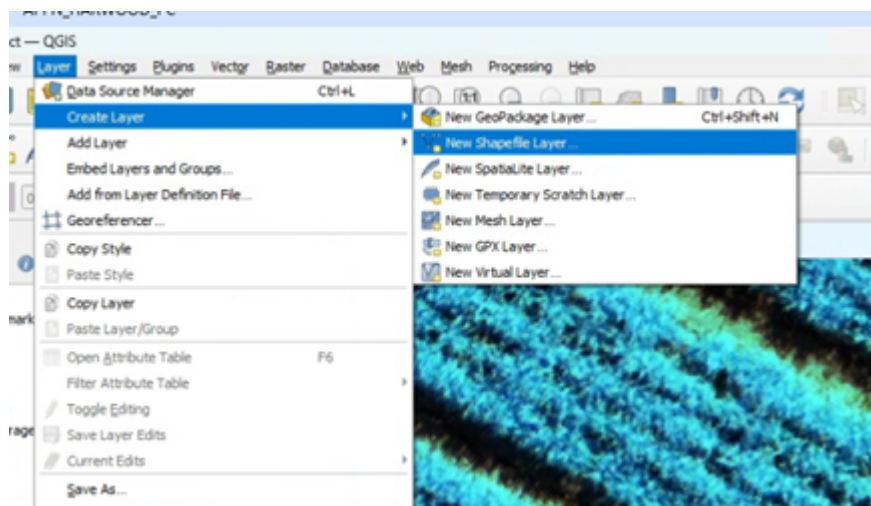


Figure: The VNIR is set to 50% opacity to confirm the panels overlap with the SWIR. The information panel is open on the SWIR to confirm the CRS (EPSG:7855 – GDA2020 / MGA zone 55). This CRS is for Narrabri NSW where this CALViS flight was collected.

2. Create the Vector Layer **Important**

TODO: Richard please check and update this section.

1. Navigate to **Layer → Create Layer → New GeoPackage Layer...** for GeoJSON output, or **New Shapefile Layer...** if producing a shapefile. Either way you will set the output **File Name** with the appropriate extension (.geojson preferred, .shp accepted).



Tip

To save directly as GeoJSON you can also create a temporary scratch layer and then **Export** → **Save Features As...** → **GeoJSON** in step 5.

2. Set the **File Name** to `QC_ELM_Panels.geojson` (see the naming conventions table for the correct name given your use case).
3. Set the **Geometry type** to *Polygon*.
4. Set the **CRS** to match your dataset.
5. Add a single field — `Panel_ref`:
 - Select the pre-filled `id` field in the Fields list and click **Remove Field** (bottom right) to remove it.
 - Set `Panel_ref` as an **Integer** between 0–100. It is the percentage reflectance (e.g. 11, 30, 56, 82 for GRYFN panels).

Important

You need to click **Add to Fields List** once you populate "New Field".

New Shapefile Layer

File name: QC_ELM_Panels.shp

File encoding: UTF-8

Geometry type: Polygon

Additional dimensions: None (selected), Z (+ M values), M values

Project CRS: EPSG:7855 - GDA2020 / MGA zone 55

New Field

Name: Panel_ref

Type: Integer (32 bit)

Length: 10 Precision: 0

Add to Fields List

Fields List

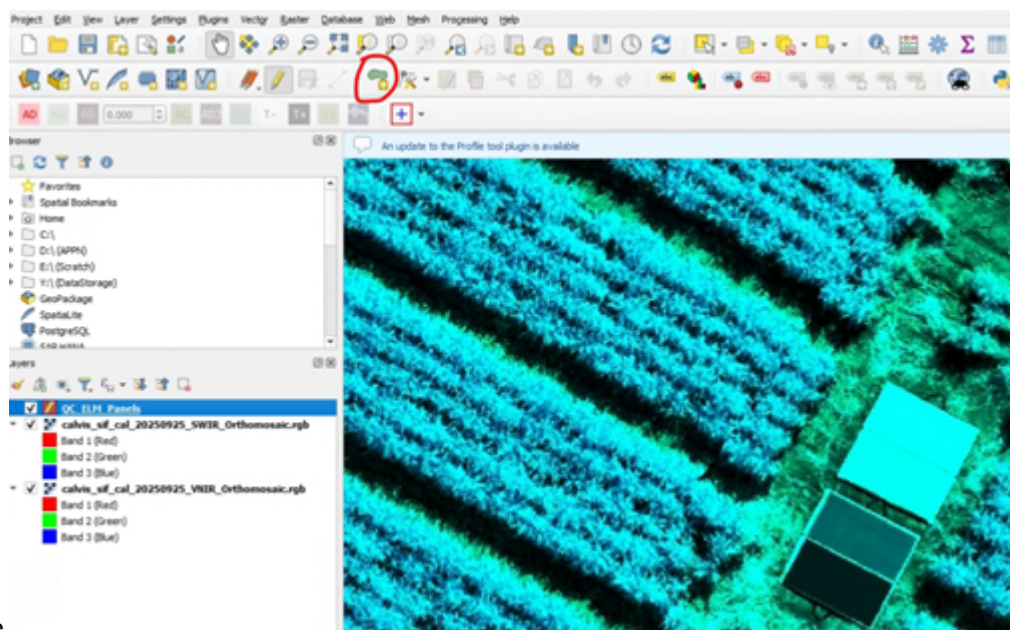
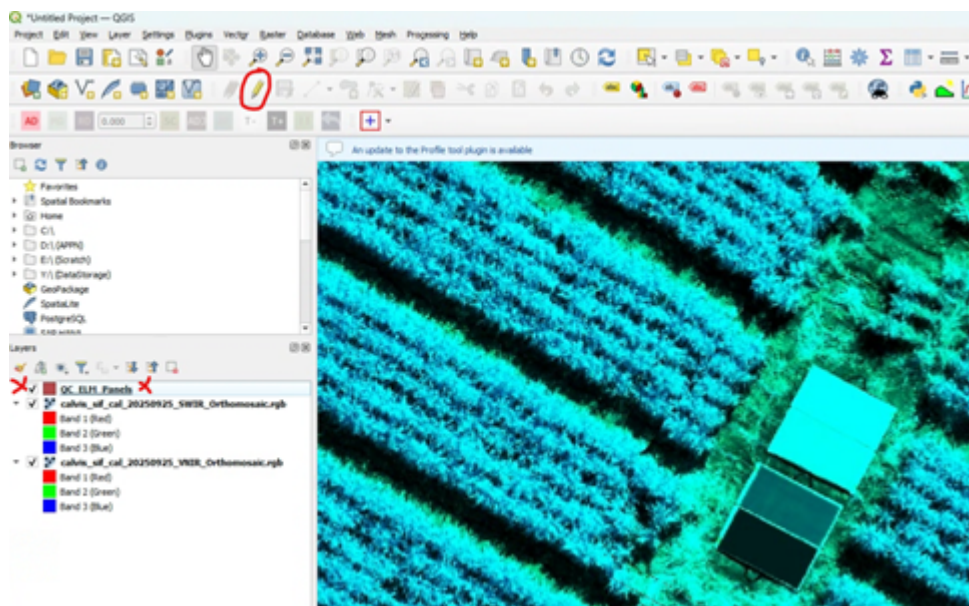
Name	Type	Length	Precision
Panel_ref	Integer	10	

Remove Field

OK Cancel Help

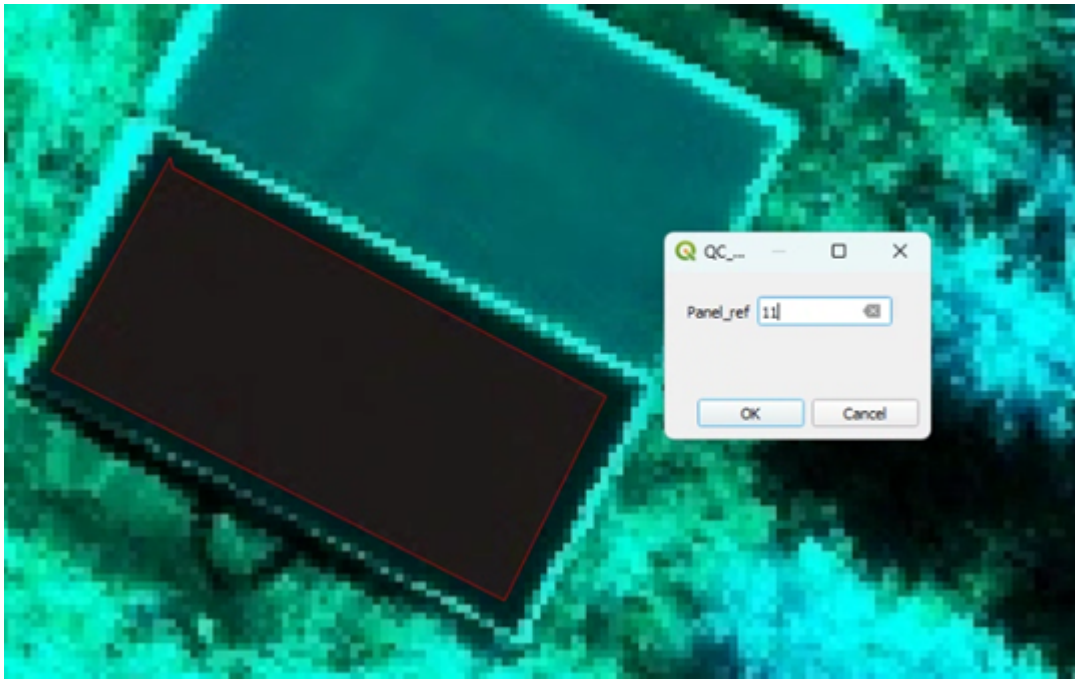
3. Draw Boxes over the Panels The boxes should be polygons drawn inside the panel. The points should be drawn to cover as much of the panel as possible without including edge effects (2–3 pixels from the edge of the panel). If there are gaps or missing values within the panel, they should be included.

1. Select **QC_ELM_Panels** in the **Layers** menu.
2. Click the pencil icon (**Toggle Editing**).



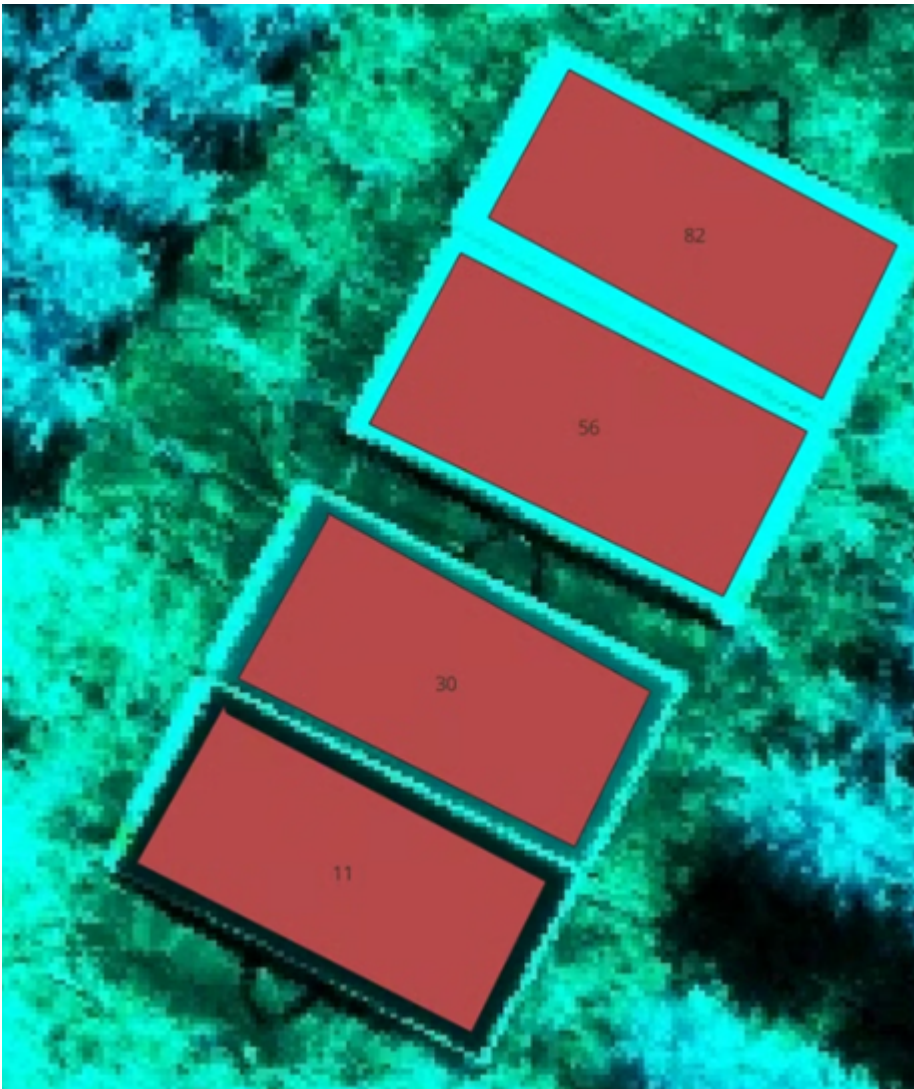
3. Click **Add Polygon Feature**

4. Click four points around a specific reflectance panel, then right-click to close the polygon. Repeat for all panels (e.g. 11, 30, 56, 82).



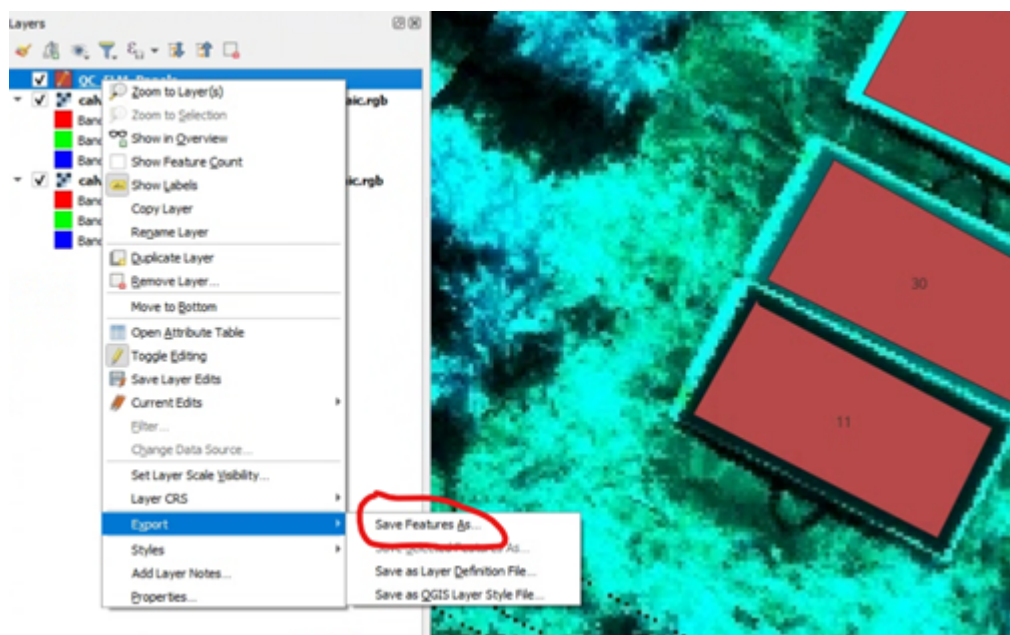
4. Sanity Check Labels

1. Double-click QC_ELM_Panels in the **Layers** menu.
2. In **Layer Properties**, click **Labels** and select **Single Labels**.

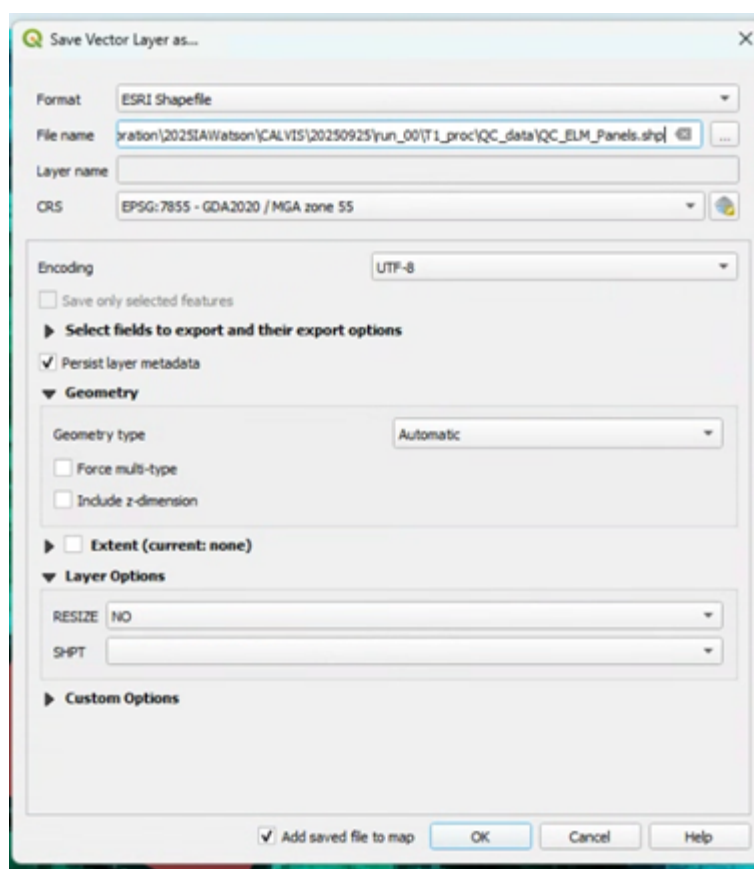


5. Save the File

1. Right-click QC_ELM_Panels in the **Layers** menu → **Export** → **Save Features As**.



2. Set the **Format** to *GeoJSON* (preferred) — or *ESRI Shapefile* if required. Make sure the **File Name** is correct and in the right folder (click the three dots to navigate). Double-check the **CRS**.



Extracting Pixels into a Table

Arden Burrell has made a Python script that can go through the APPN standard folder structure, extract the values into a table, and save that as a `.csv` or `.parquet` file automatically. The code is available from:

https://github.com/ArdenB/APPN_GenricFileStorage

- **Script:** Code/DS02_DatasetQA/QA00_ELMvalidation.py
- **README:** Code/DS02_DatasetQA/README.md

If nodes choose to extract the points using other means, the tables should have the following columns:

band, wavelength, value, Panel_ref, node, project, site, sensor, date, run, panel_name, type, gpro_nu

LiDAR QC

Warning

This section is still a work in progress.